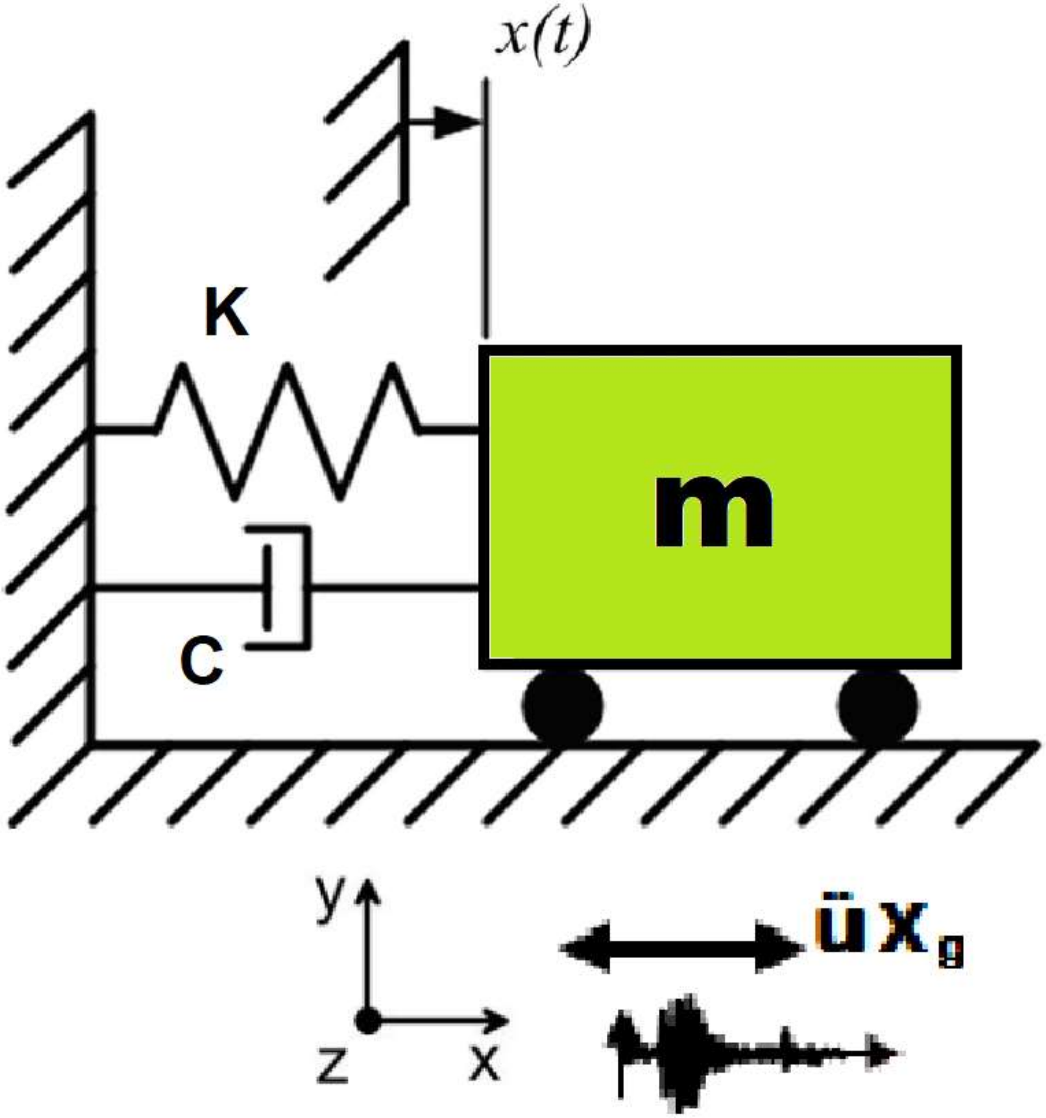
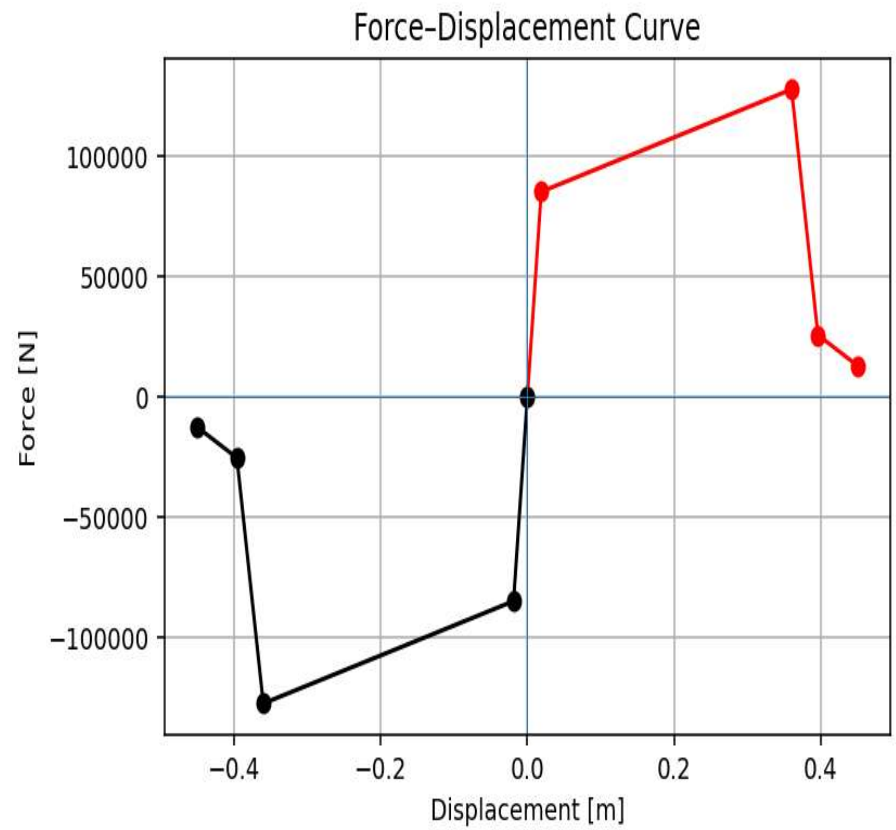
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

FRAGILITY ANALYSIS BASED ON ACCELERATION AND STRUCTURAL DUCTILITY DAMAGE INDEX AND EVALUATION OF EQULIVALENT VISCOUS DAMPING RATIO WITH INCREMENTAL DYNAMIC ANALYSIS (IDA) OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM UTILIZING 100 GROUND MOTIONS IN OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


x(t)
K
m
C
y
z
x
ü x g


Force-Displacement Curve
100000
50000
0
−50000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

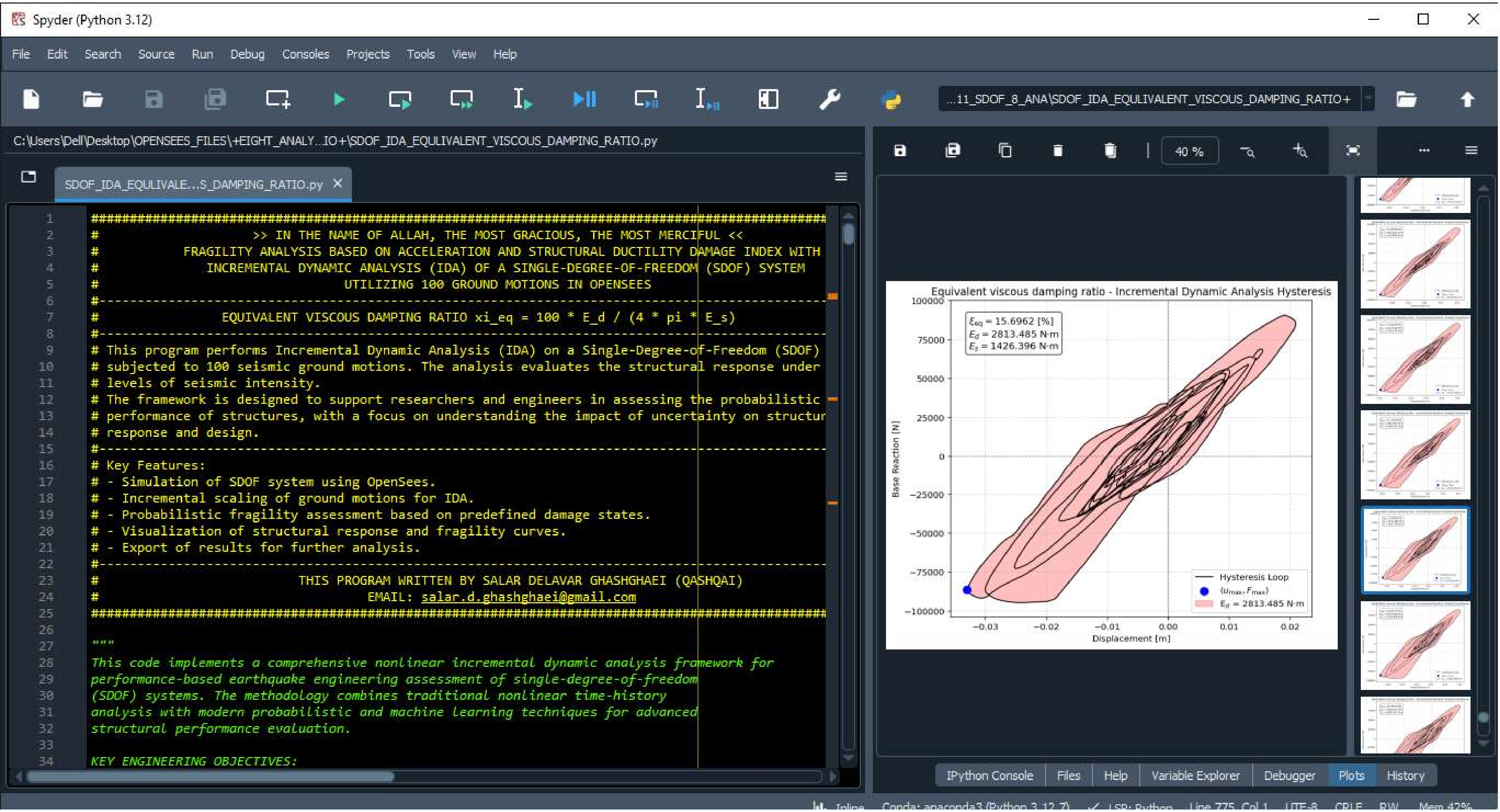
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

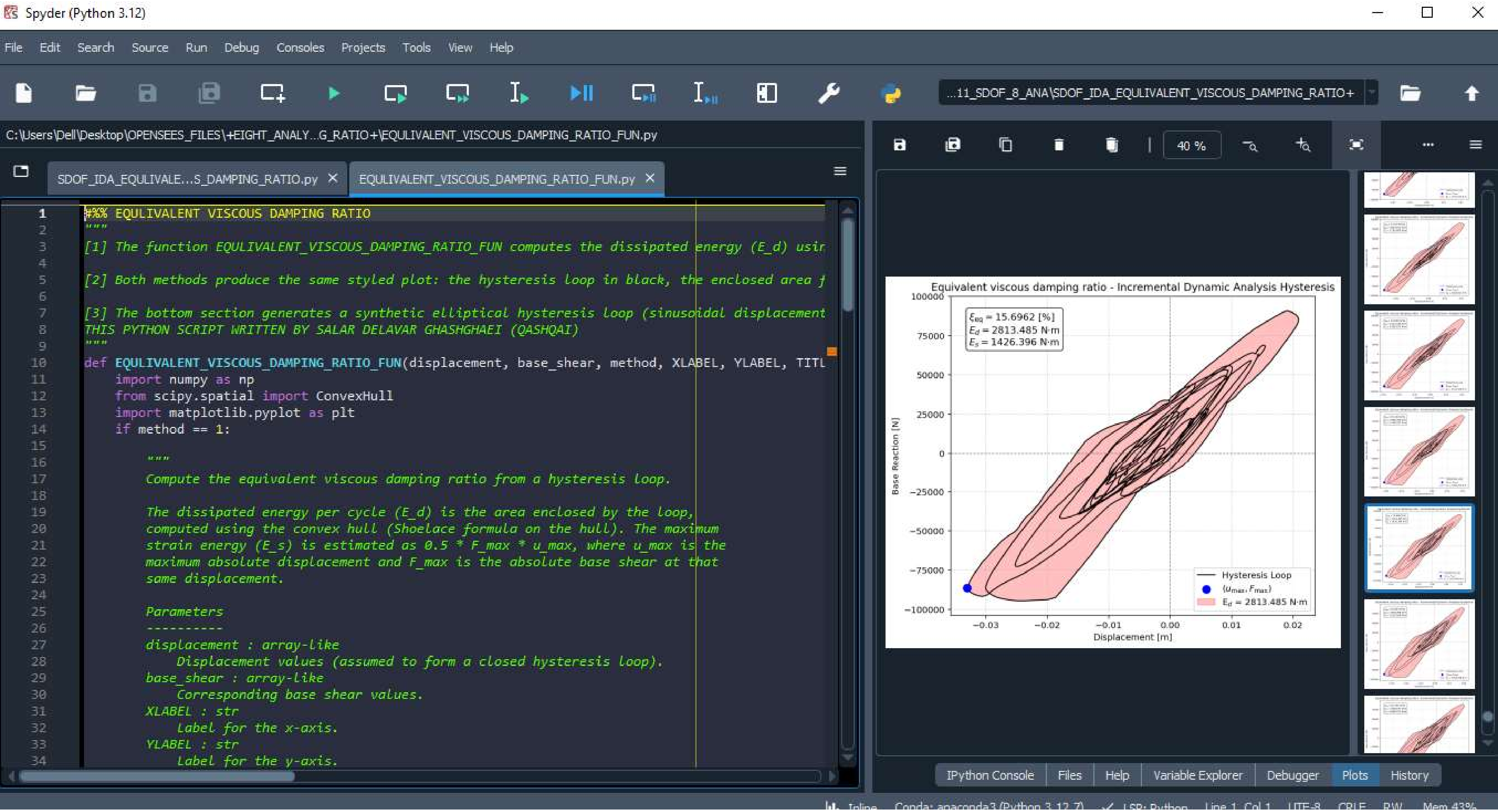
  where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{sec} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

The equivalent viscous damping ratio is then:

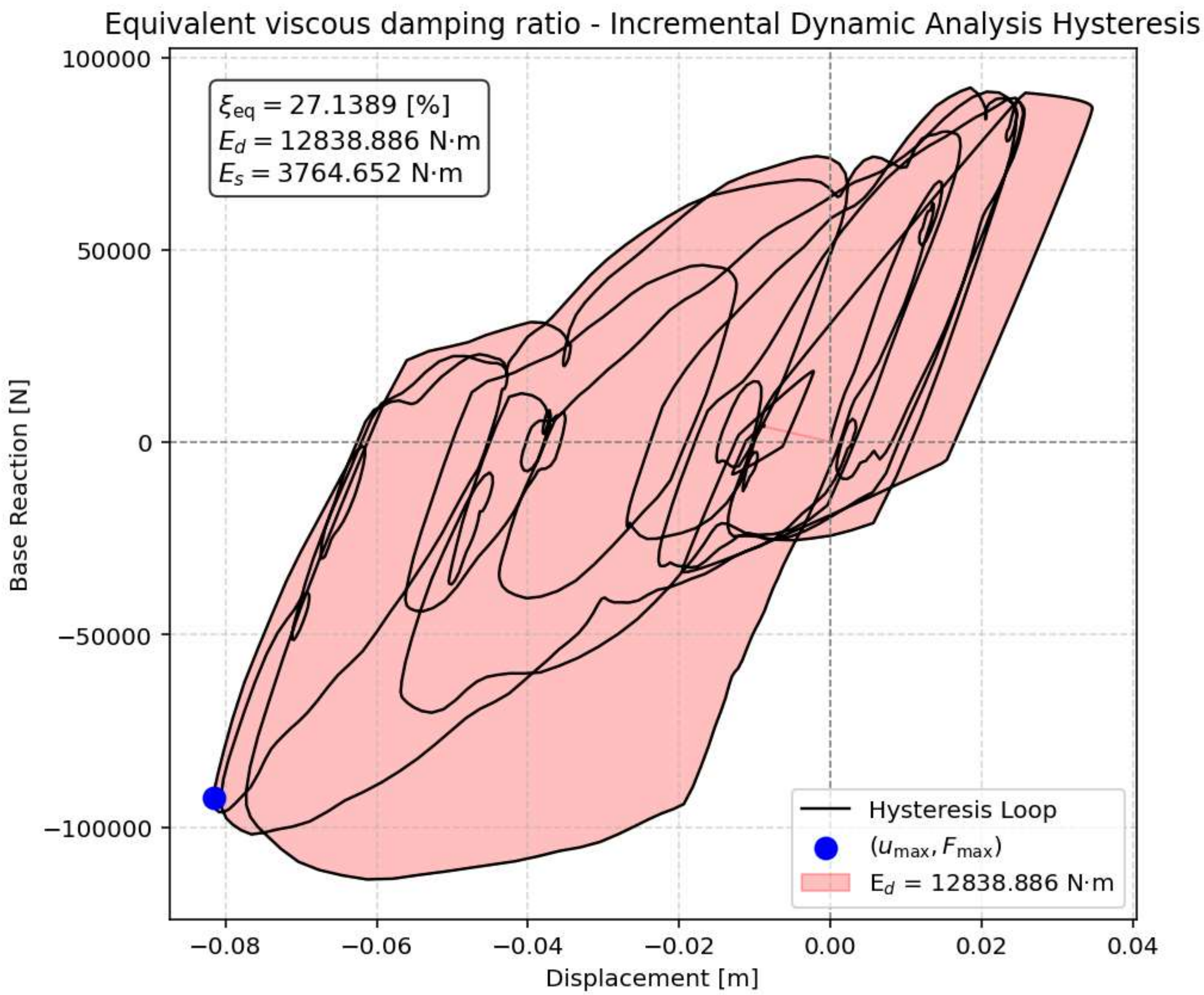
$$\xi_{eq} = \frac{100\, E_d}{4\pi\, E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.

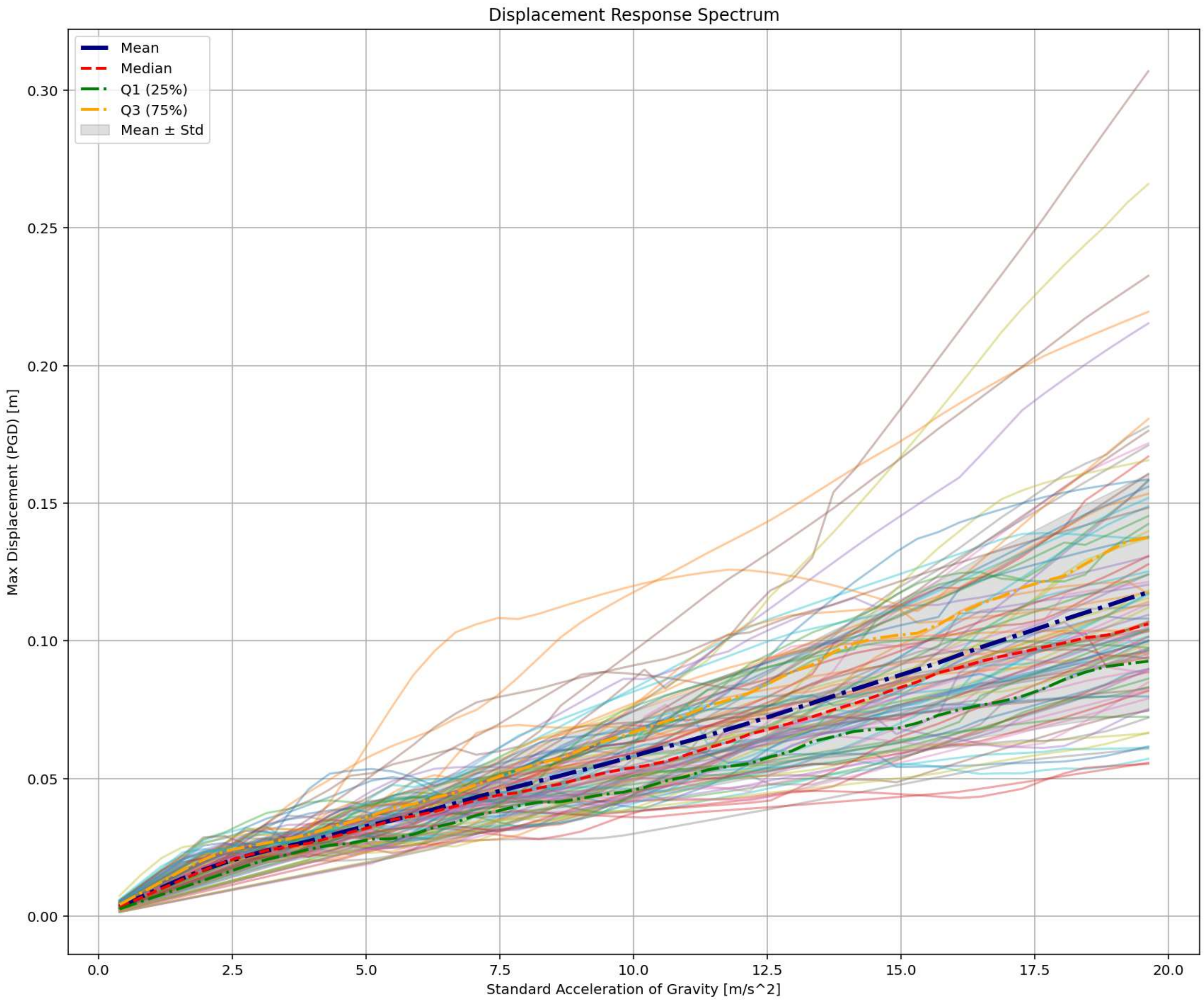

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...11_SDOF_8_ANA\SDOF_IDA_EQULIVALENT_VISCOUS_DAMPING_RATIO+
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IO+\SDOF_IDA_EQULIVALENT_VISCOUS_DAMPING_RATIO.py
SDOF_IDA_EQULIVALE...S_DAMPING_RATIO.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
FRAGILITY ANALYSIS BASED ON ACCELERATION AND STRUCTURAL DUCTILITY DAMAGE INDEX WITH
INCREMENTAL DYNAMIC ANALYSIS (IDA) OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM
UTILIZING 100 GROUND MOTIONS IN OPENSEES
EQUIVALENT VISCOUS DAMPING RATIO xi_eq = 100 * E_d / (4 * pi * E_s)
This program performs Incremental Dynamic Analysis (IDA) on a Single-Degree-of-Freedom (SDOF)
subjected to 100 seismic ground motions. The analysis evaluates the structural response under
levels of seismic intensity.
The framework is designed to support researchers and engineers in assessing the probabilistic
performance of structures, with a focus on understanding the impact of uncertainty on structur
response and design.
Key Features:
- Simulation of SDOF system using OpenSees.
- Incremental scaling of ground motions for IDA.
- Probabilistic fragility assessment based on predefined damage states.
- Visualization of structural response and fragility curves.
- Export of results for further analysis.
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
This code implements a comprehensive nonlinear incremental dynamic analysis framework for
performance-based earthquake engineering assessment of single-degree-of-freedom
(SDOF) systems. The methodology combines traditional nonlinear time-history
analysis with modern probabilistic and machine learning techniques for advanced
structural performance evaluation.
KEY ENGINEERING OBJECTIVES:
40 %
Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
ξeq = 15.6962 [%]
Ed = 2813.485 N·m
Es = 1426.396 N·m
Base Reaction [N]
Displacement [m]
Hysteresis Loop
(umax, Fmax)
Ed = 2813.485 N·m
IPython Console Files Help Variable Explorer Debugger Plots History
Conda: anaconda3 (Python 3.12.7) LSP: Python Line 775, Col 1 UTF-8 CRLF RW Mem 42%

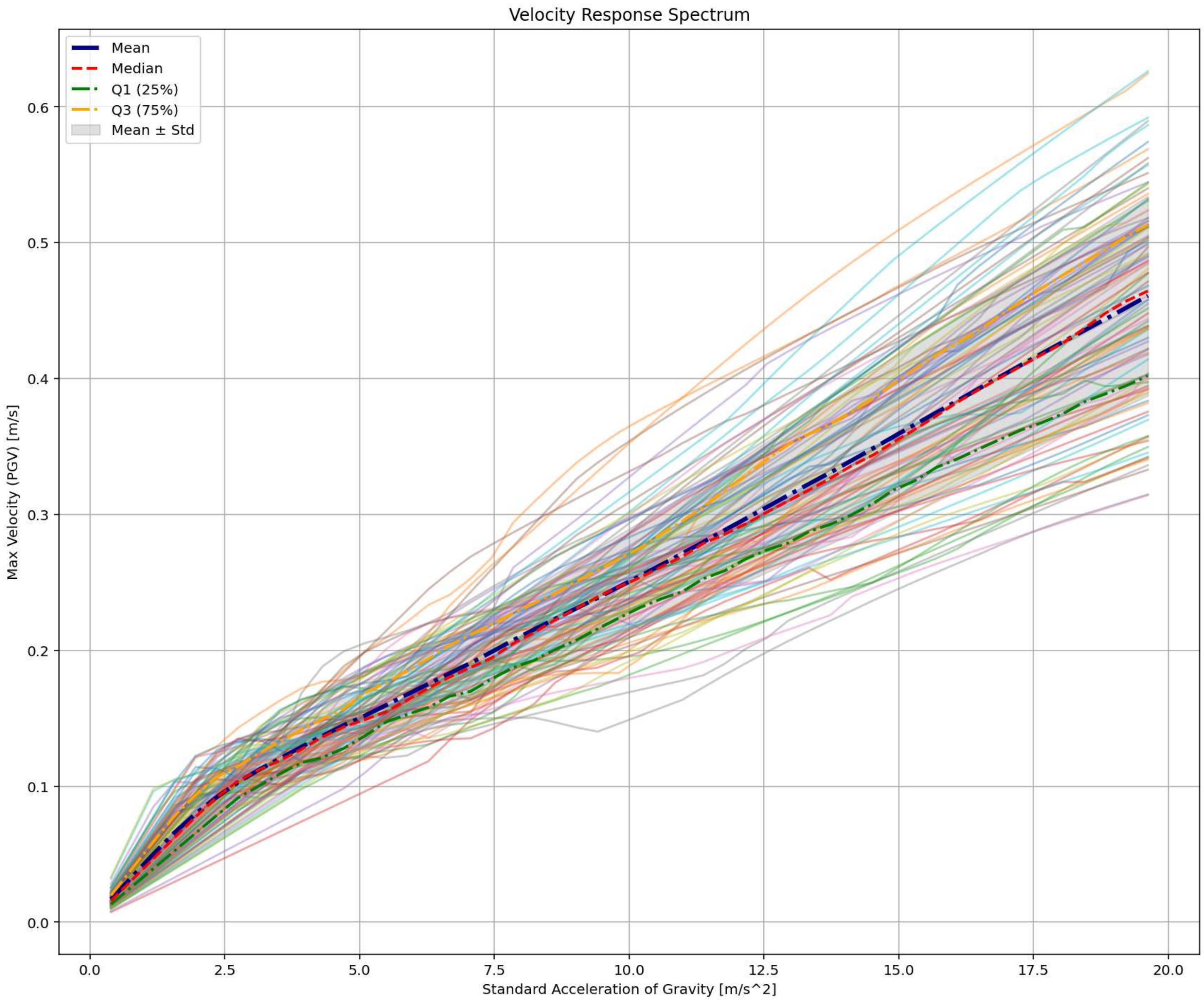


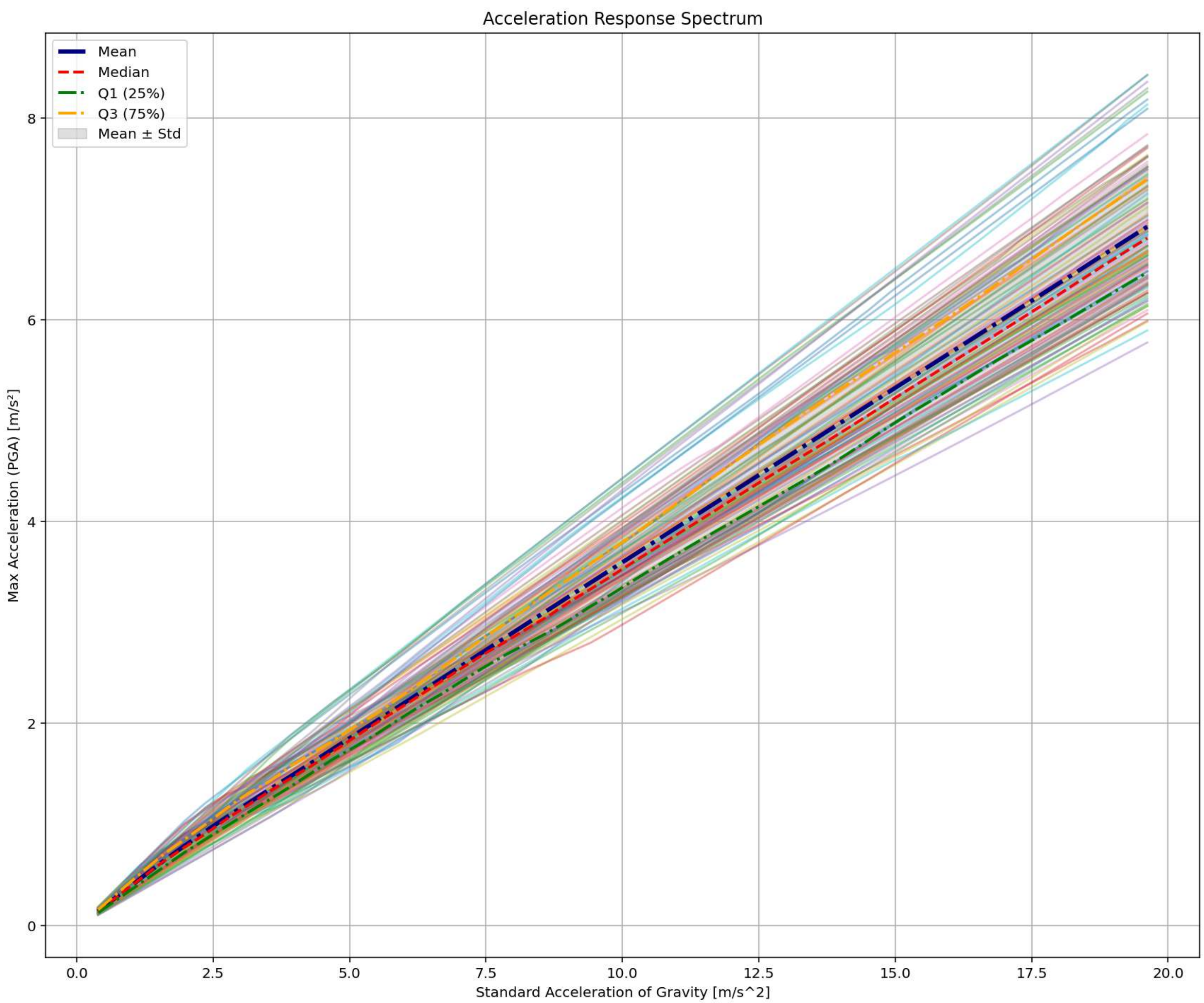
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...11_SDOF_8_ANA\SDOF_IDA_EQULIVALENT_VISCOUS_DAMPING_RATIO+
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...G_RATIO+\EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py
SDOF_IDA_EQULIVALE...S_DAMPING_RATIO.py
EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py
#%% EQULIVALENT VISCOUS DAMPING RATIO
"""
[1] The function EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN computes the dissipated energy (E_d) usir
[2] Both methods produce the same styled plot: the hysteresis loop in black, the enclosed area f
[3] The bottom section generates a synthetic elliptical hysteresis loop (sinusoidal displacement
THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
def EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN(displacement, base_shear, method, XLABEL, YLABEL, TITL
    import numpy as np
    from scipy.spatial import ConvexHull
    import matplotlib.pyplot as plt
    if method == 1:
        """
        Compute the equivalent viscous damping ratio from a hysteresis loop.
        The dissipated energy per cycle (E_d) is the area enclosed by the loop,
        computed using the convex hull (Shoelace formula on the hull). The maximum
        strain energy (E_s) is estimated as 0.5 * F_max * u_max, where u_max is the
        maximum absolute displacement and F_max is the absolute base shear at that
        same displacement.
        Parameters
        ----
        displacement : array-like
            Displacement values (assumed to form a closed hysteresis loop).
        base_shear : array-like
            Corresponding base shear values.
        XLABEL : str
            Label for the x-axis.
        YLABEL : str
            Label for the y-axis.
40 %
Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
ξeq = 15.6962 [%]
Ed = 2813.485 N·m
Es = 1426.396 N·m
Hysteresis Loop
(umax, Fmax)
Ed = 2813.485 N·m
Base Reaction [N]
Displacement [m]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%

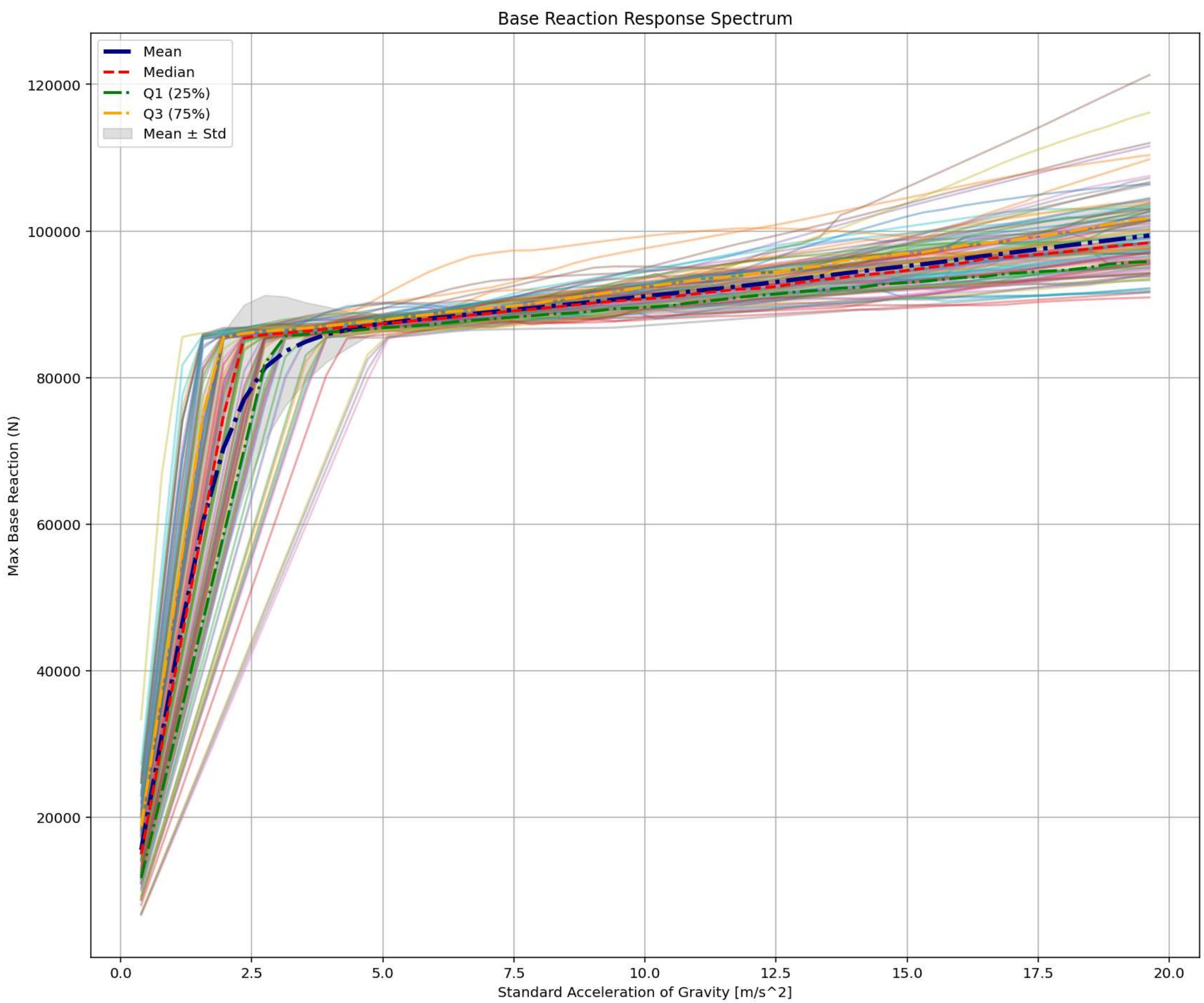


Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
$\xi_{eq} = 27.1389$ [%]
$E_d = 12838.886$ N·m
$E_s = 3764.652$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E_d = 12838.886 N·m
Base Reaction [N]
Displacement [m]
100000
50000
0
−50000
−100000
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04

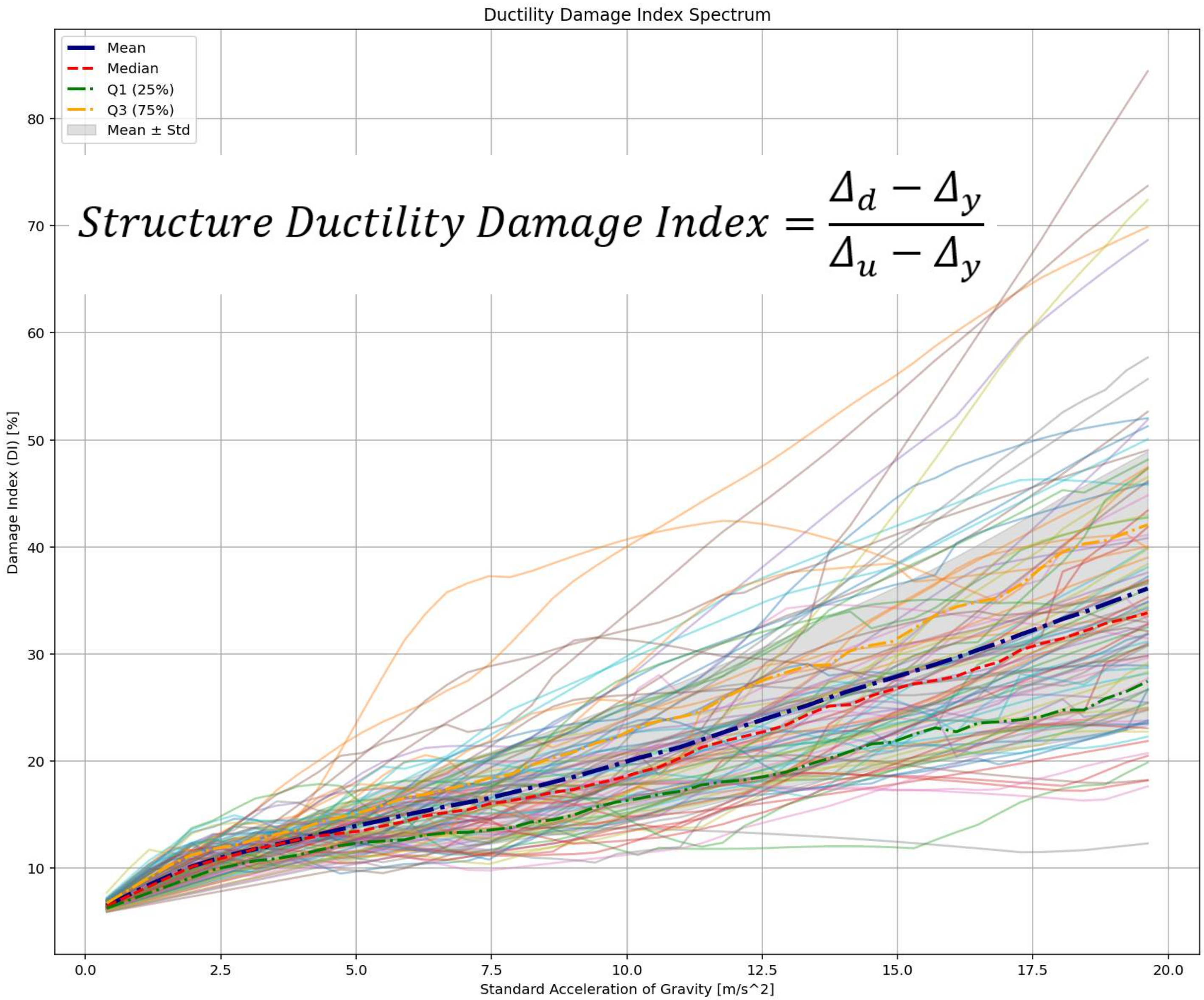

Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Displacement (PGD) [m]
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Standard Acceleration of Gravity [m/s^2]

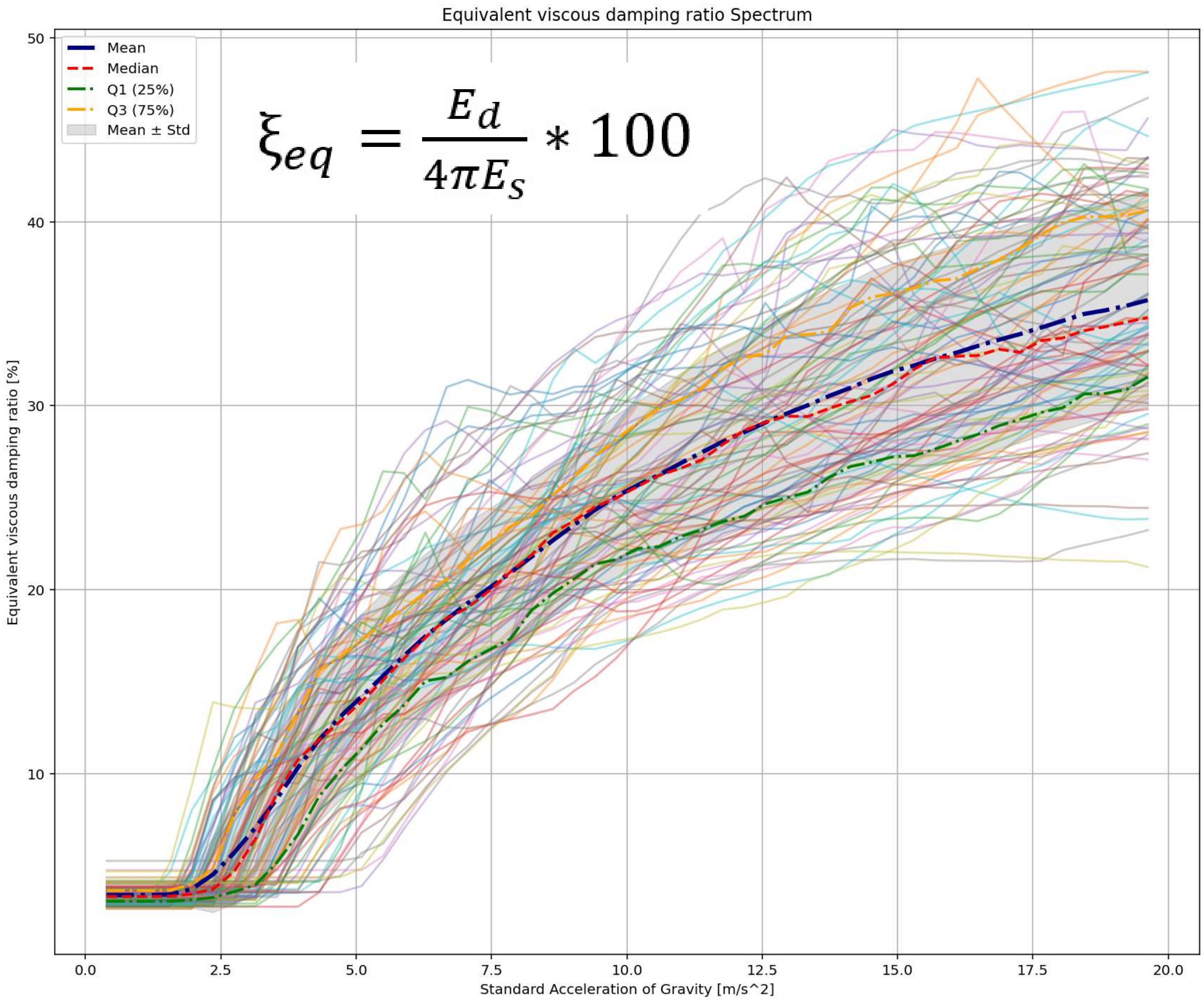

Velocity Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Velocity (PGV) [m/s]
Standard Acceleration of Gravity [m/s^2]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

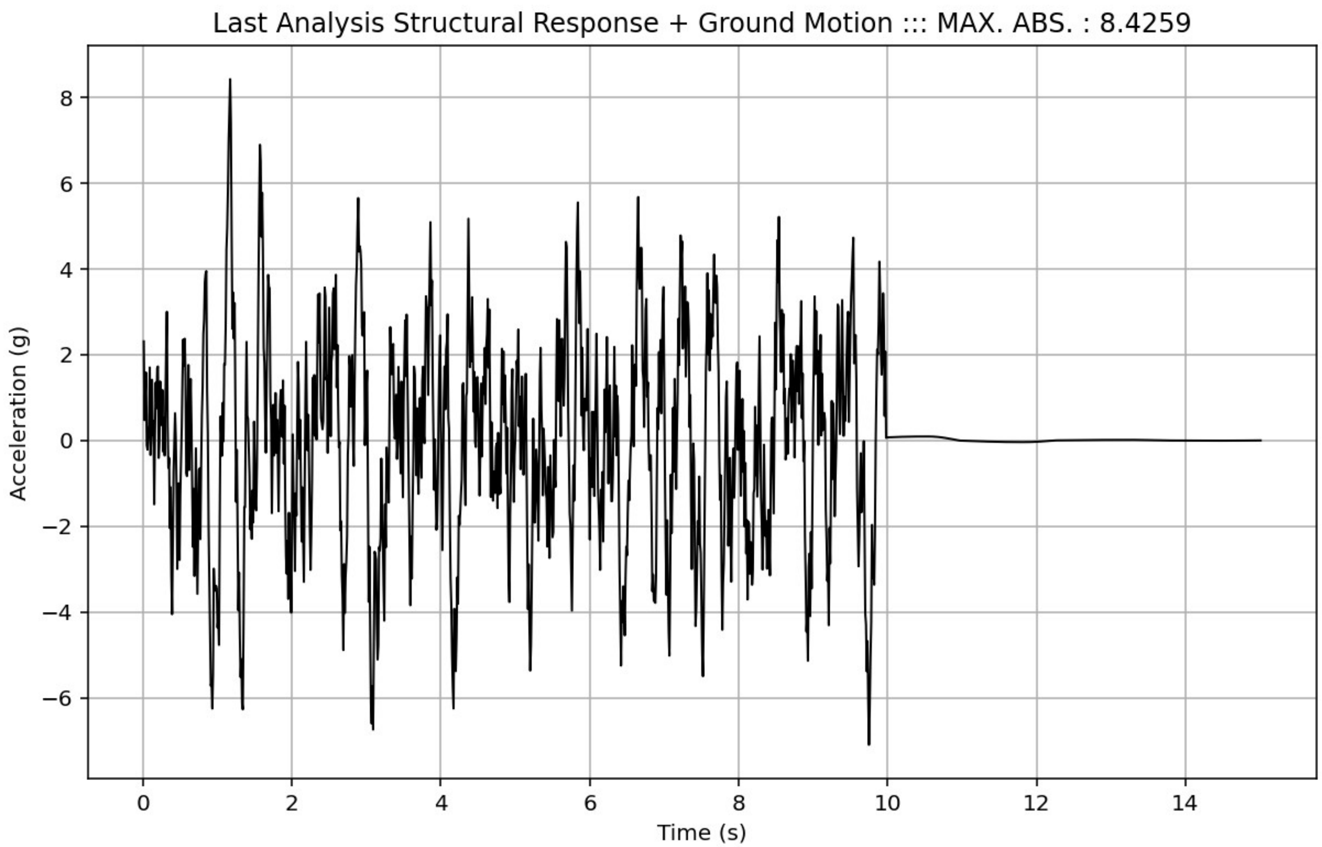

Acceleration Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Acceleration (PGA) [m/s²]
Standard Acceleration of Gravity [m/s^2]
0
2
4
6
8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

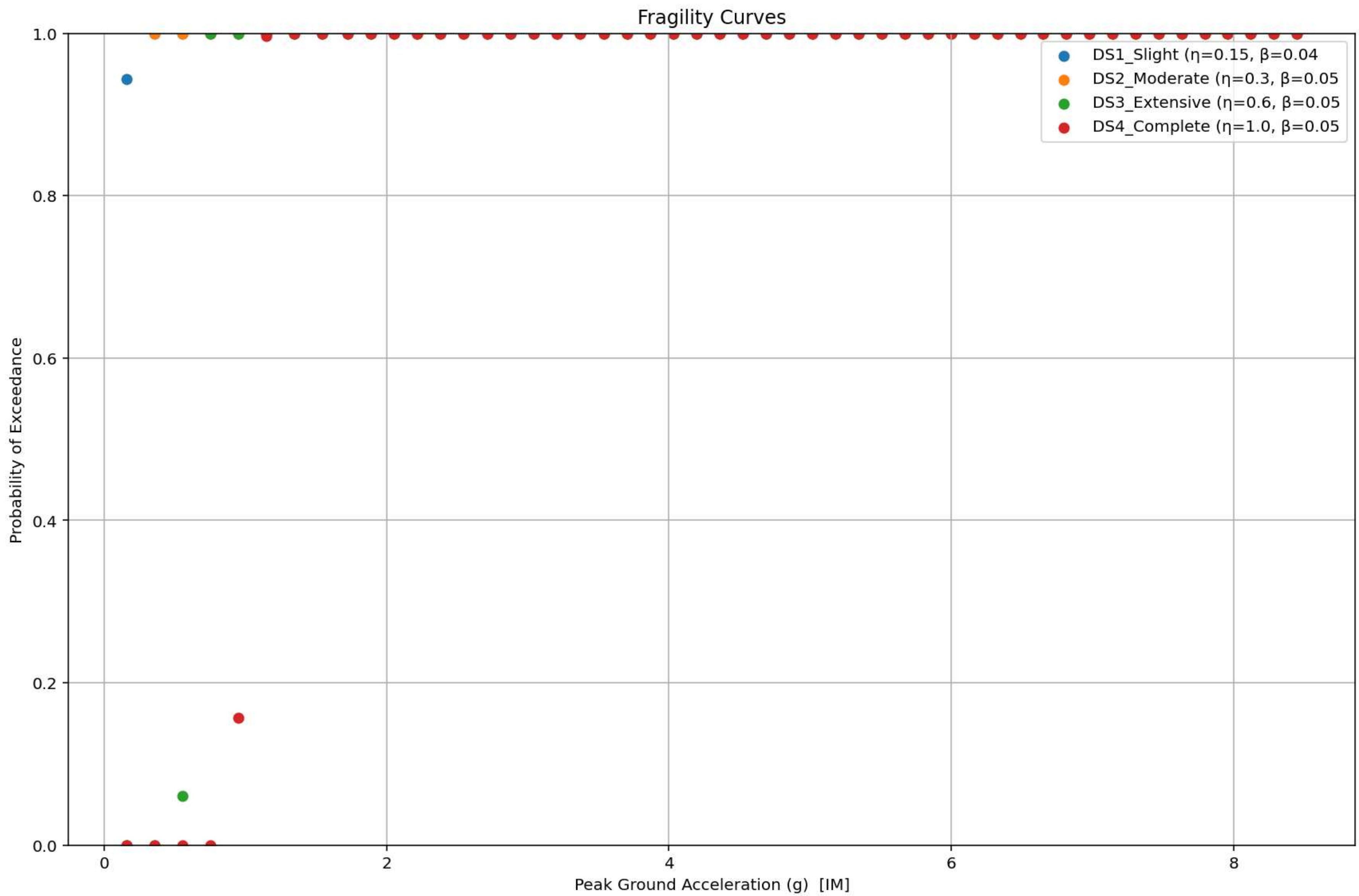


Base Reaction Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
120000
100000
80000
60000
40000
20000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Standard Acceleration of Gravity [m/s^2]

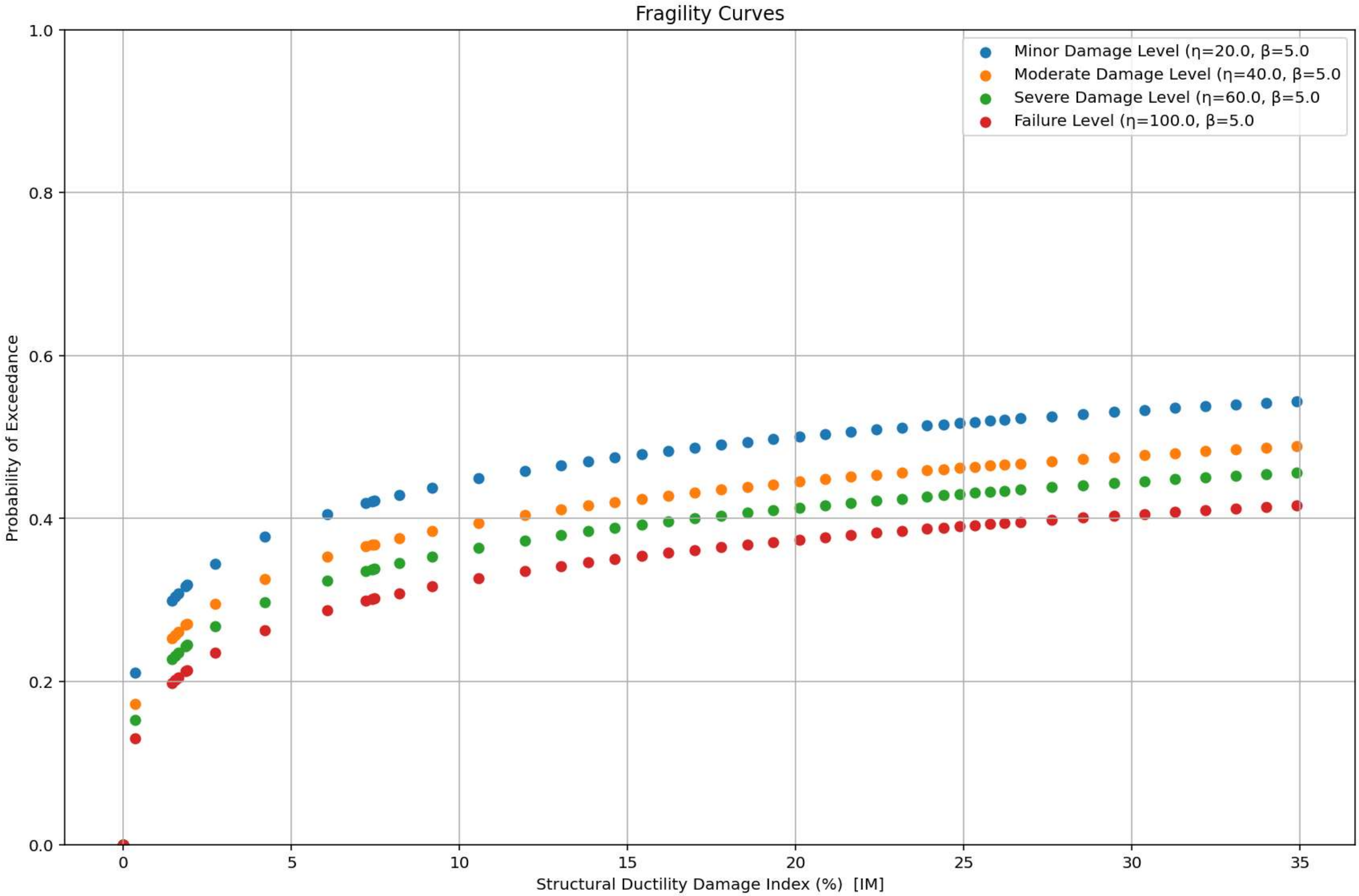

Ductility Damage Index Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$
Damage Index (DI) [%]
Standard Acceleration of Gravity [m/s^2]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10
20
30
40
50
60
70
80

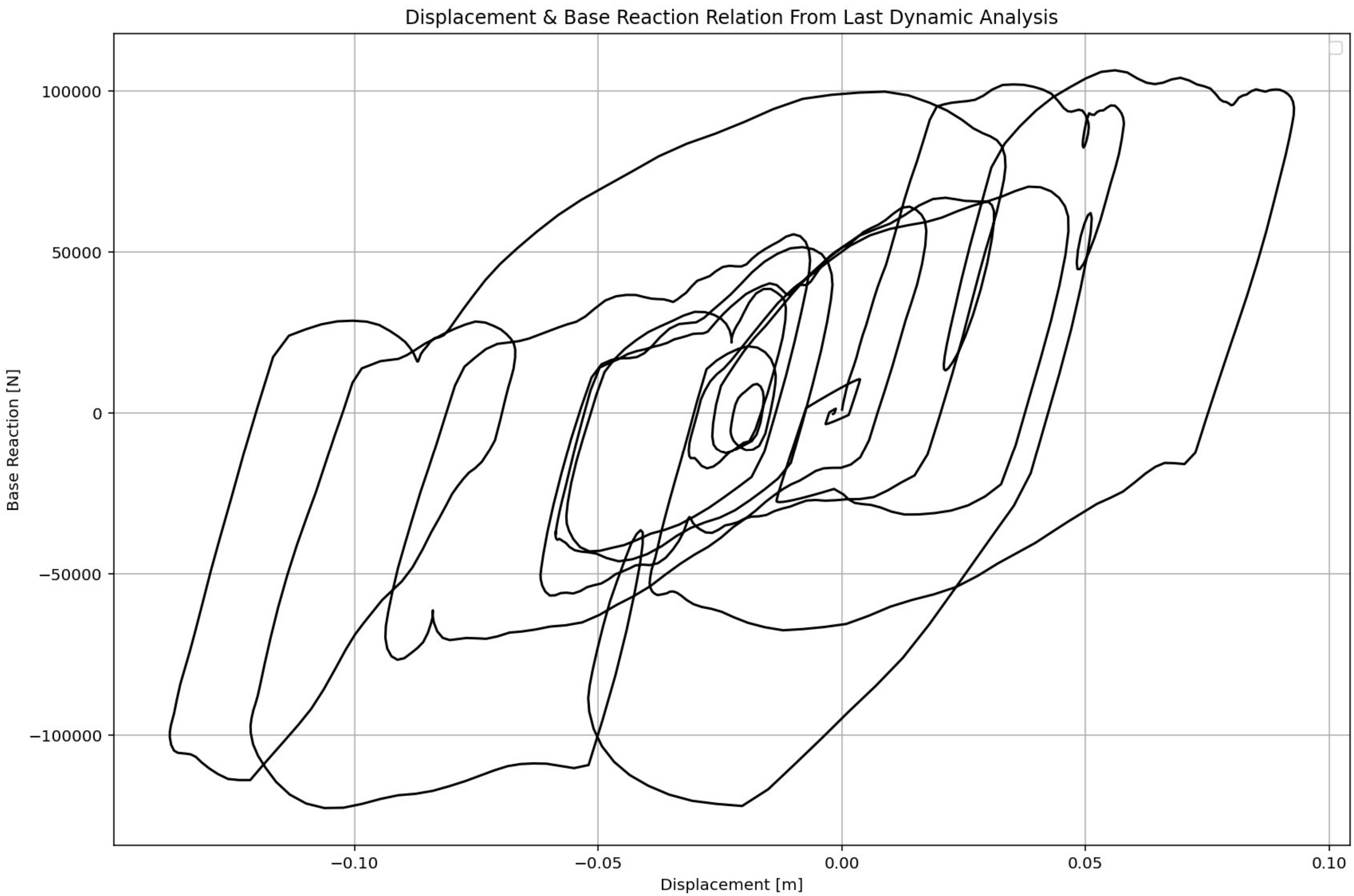

Equivalent viscous damping ratio Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
$$\xi_{eq} = \frac{E_d}{4\pi E_S} * 100$$
Equivalent viscous damping ratio [%]
Standard Acceleration of Gravity [m/s^2]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10
20
30
40
50


Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 8.4259
Acceleration (g)
Time (s)
8
6
4
2
0
−2
−4
−6
0
2
4
6
8
10
12
14

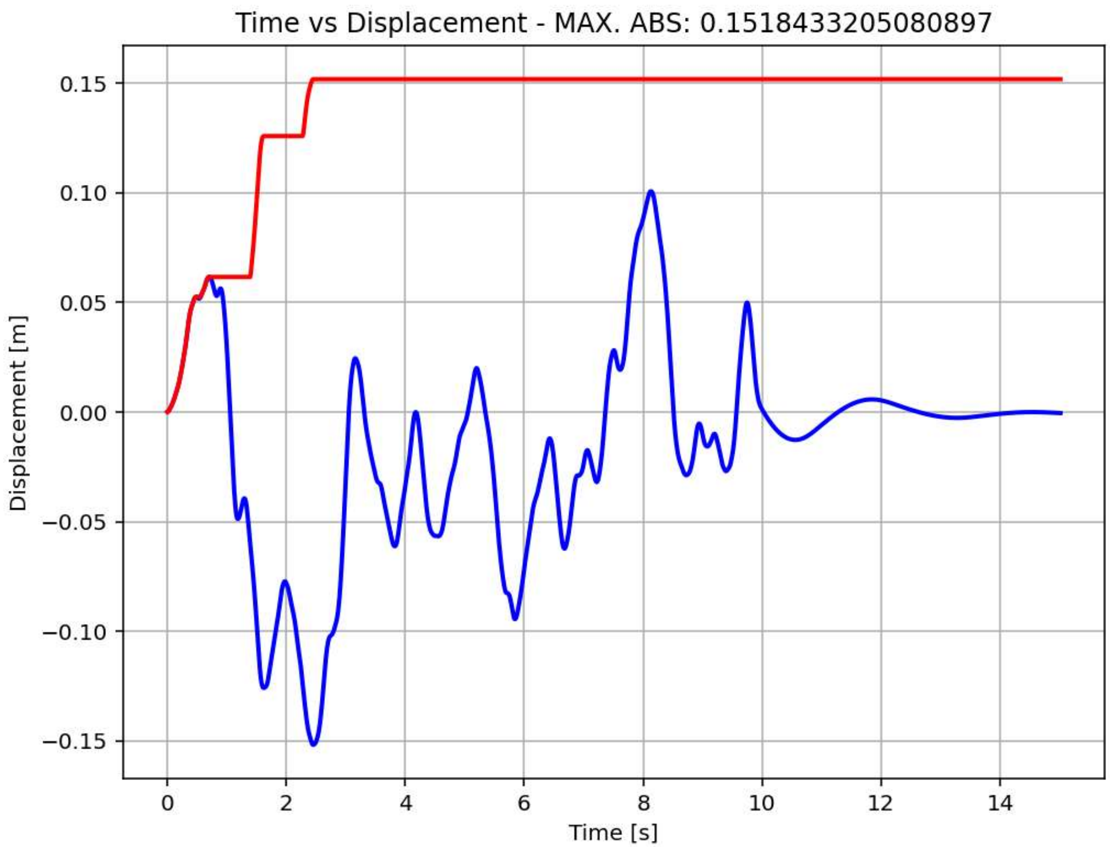


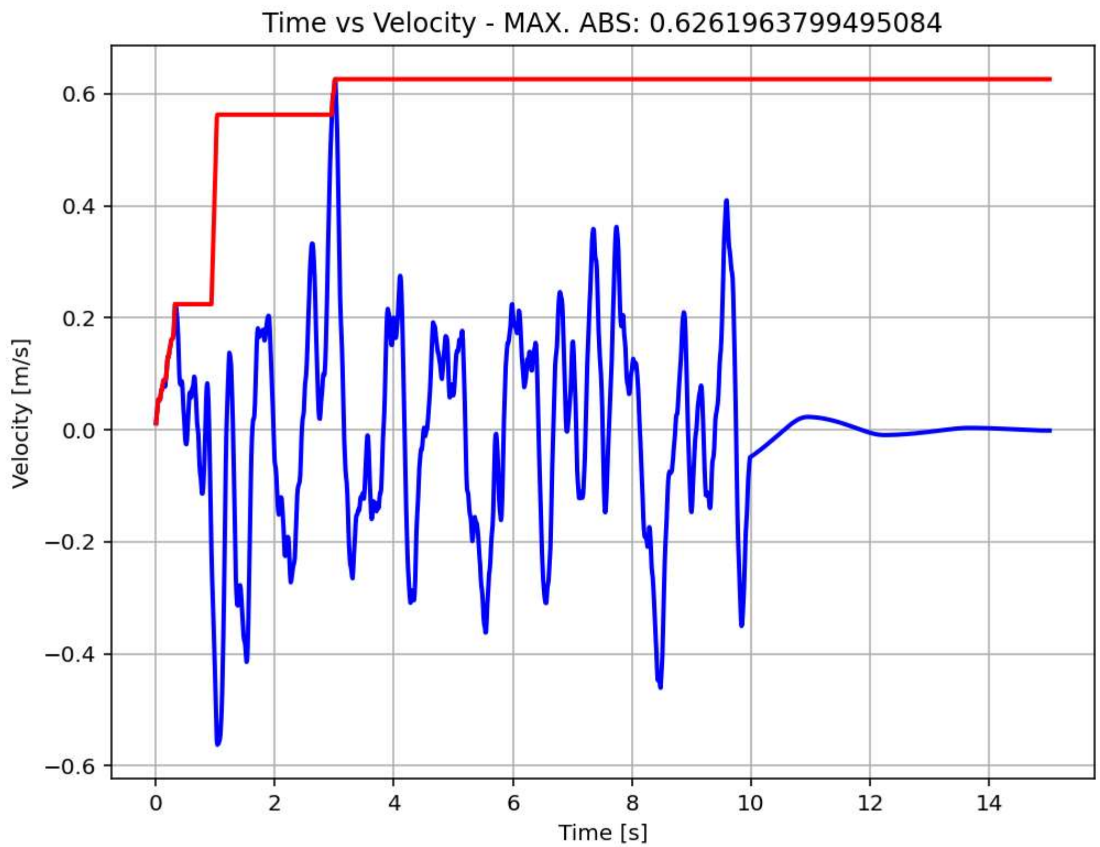
Fragility Curves
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Probability of Exceedance
Peak Ground Acceleration (g) [IM]
DS1_Slight (η=0.15, β=0.04
DS2_Moderate (η=0.3, β=0.05
DS3_Extensive (η=0.6, β=0.05
DS4_Complete (η=1.0, β=0.05


Fragility Curves
Minor Damage Level (η=20.0, β=5.0
Moderate Damage Level (η=40.0, β=5.0
Severe Damage Level (η=60.0, β=5.0
Failure Level (η=100.0, β=5.0
Probability of Exceedance
Structural Ductility Damage Index (%) [IM]
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30
35

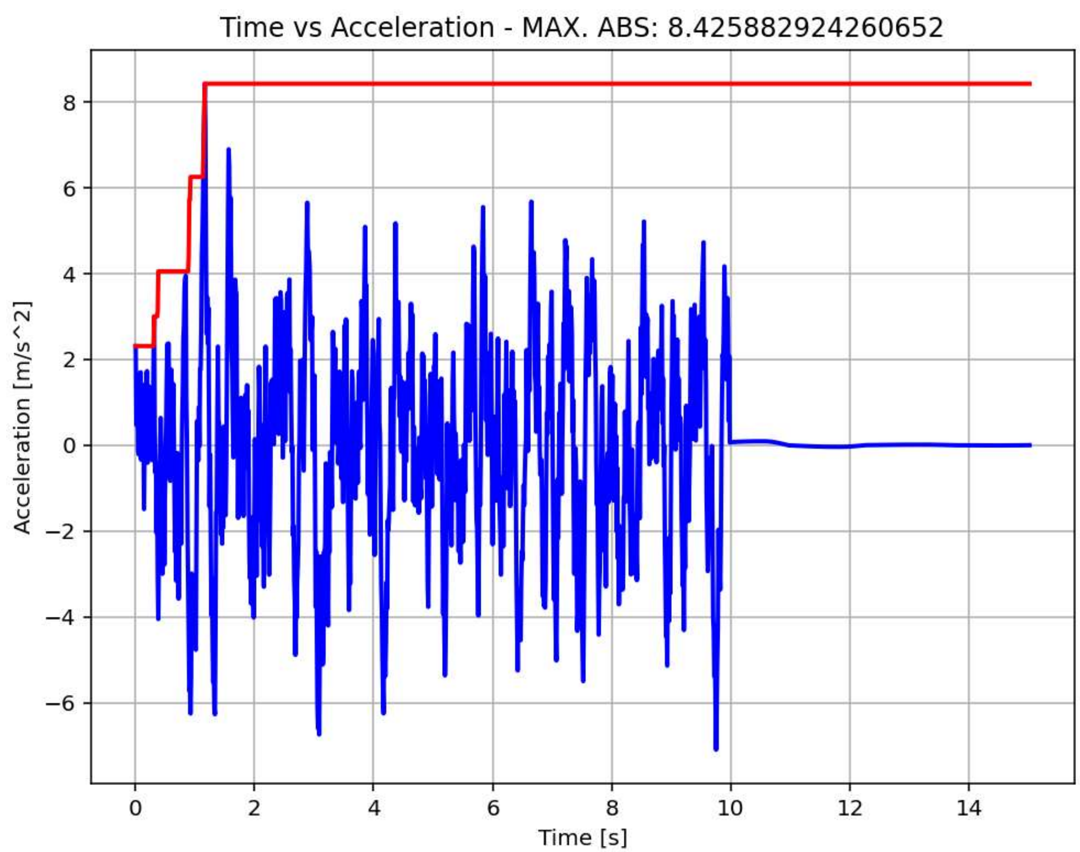


Displacement & Base Reaction Relation From Last Dynamic Analysis
Base Reaction [N]
100000
50000
0
−50000
−100000
−0.10
−0.05
0.00
0.05
0.10
Displacement [m]

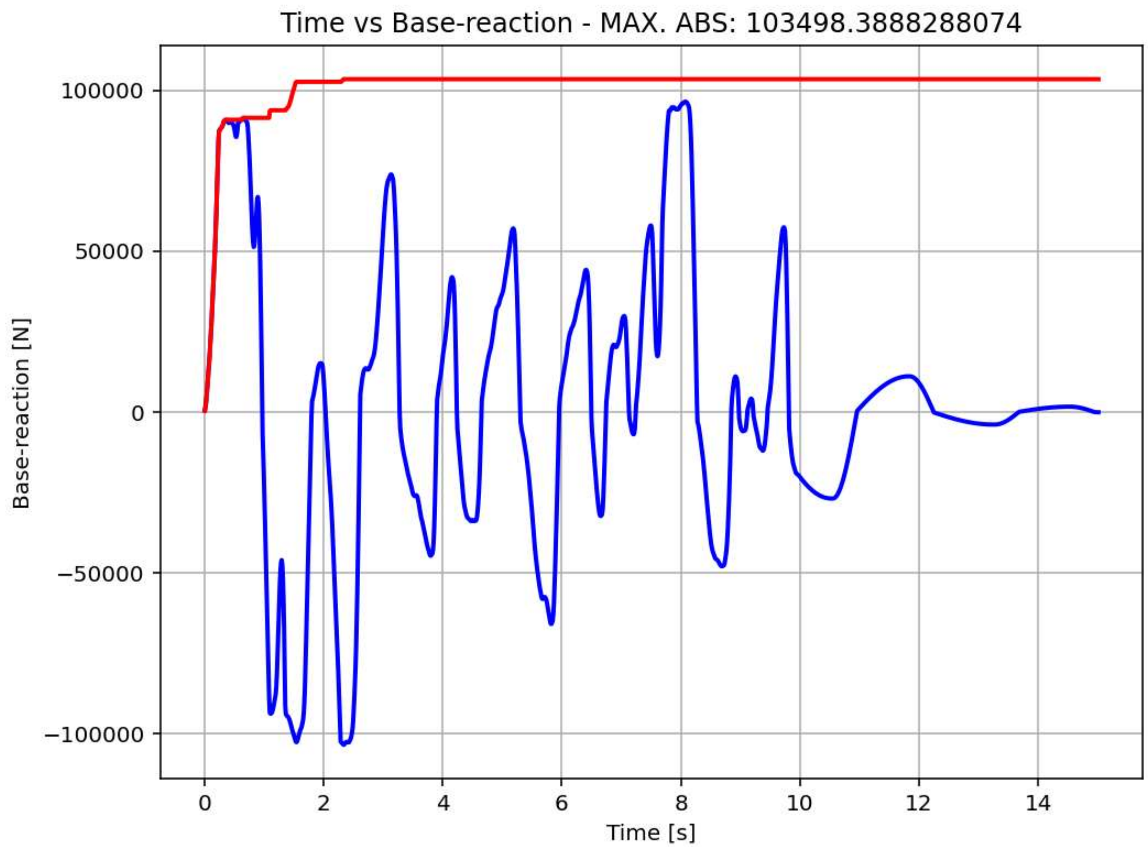

Time vs Displacement - MAX. ABS: 0.1518433205080897
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Displacement [m]
0
2
4
6
8
10
12
14
Time [s]

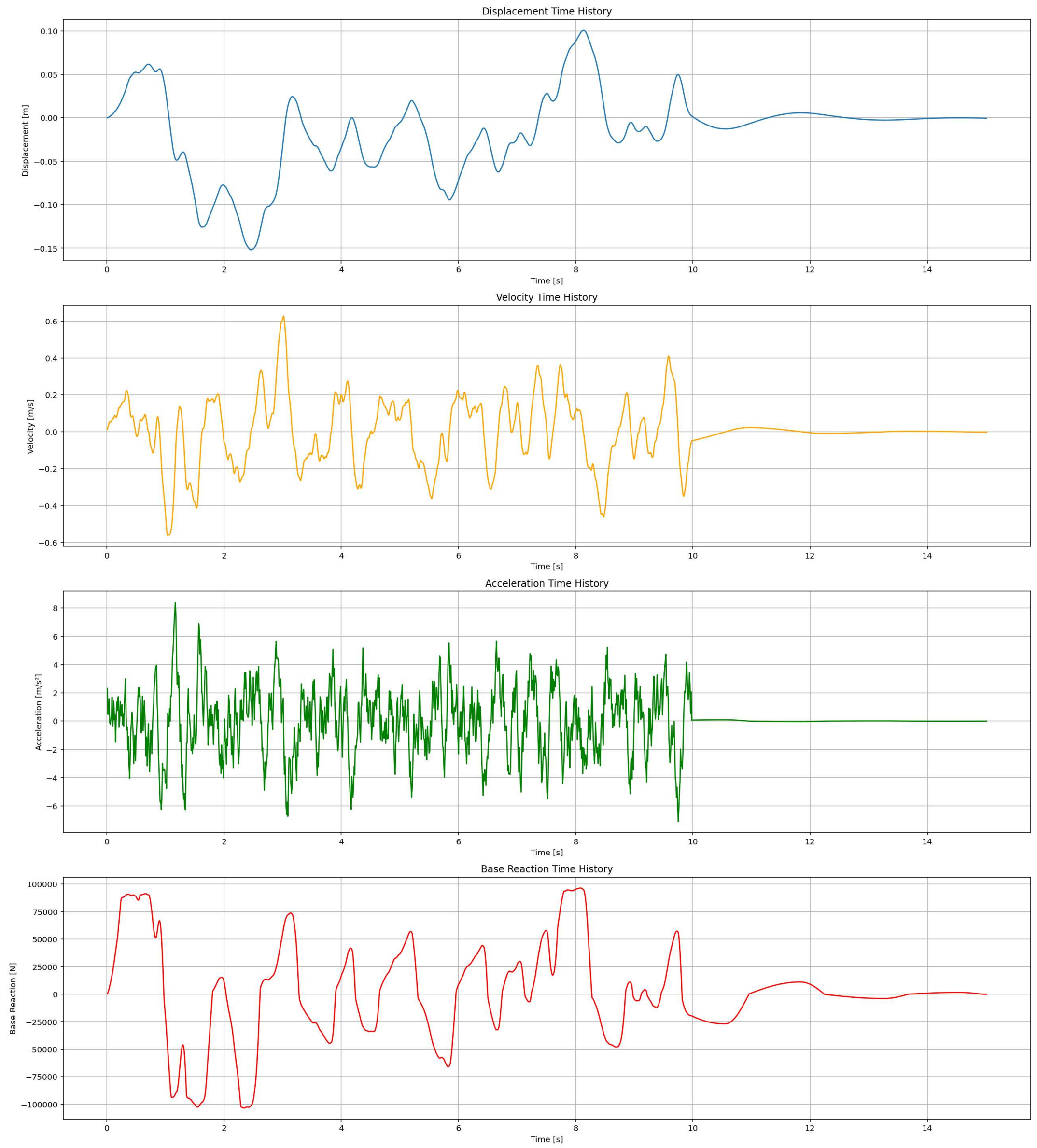


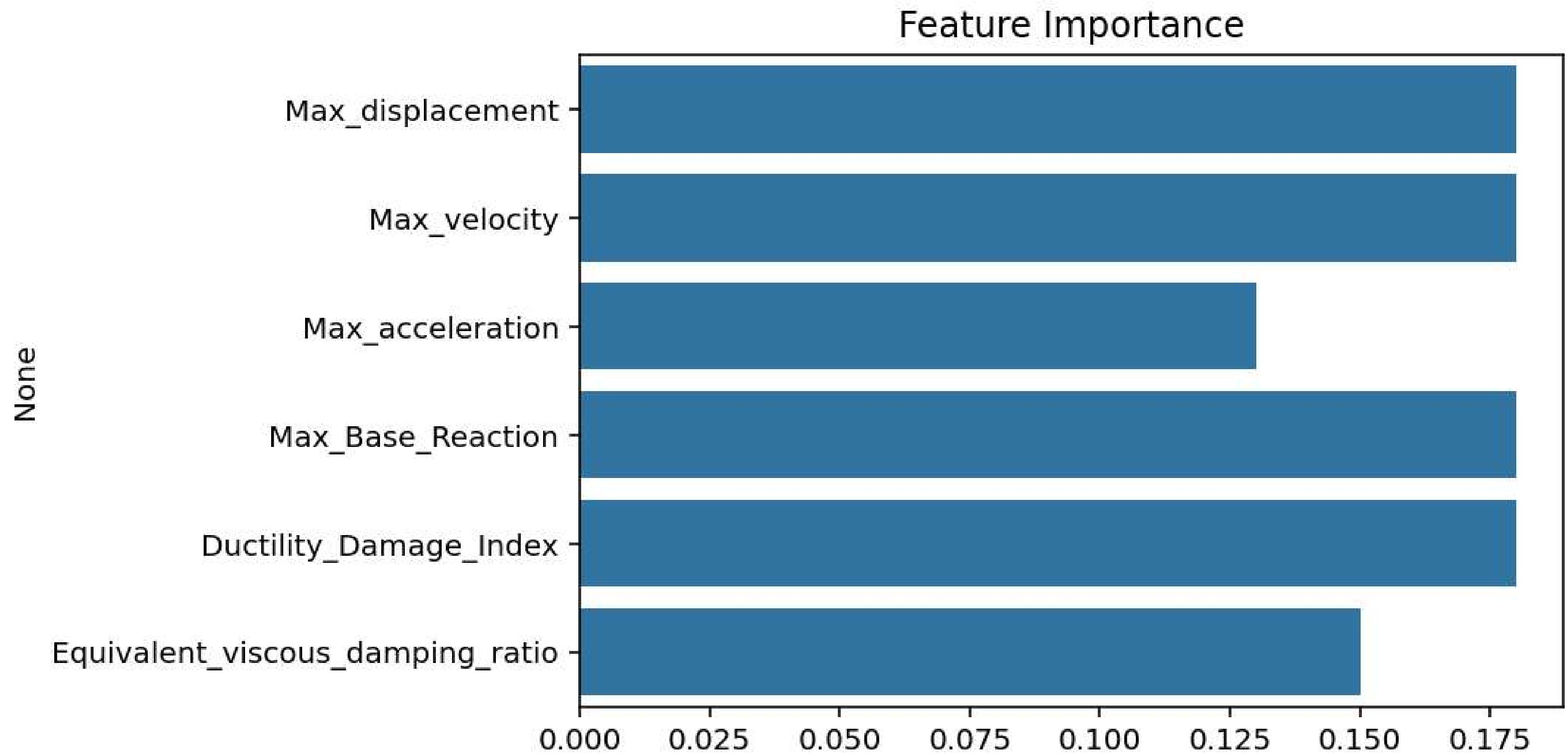
Time vs Velocity - MAX. ABS: 0.6261963799495084
Velocity [m/s]
Time [s]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
2
4
6
8
10
12
14



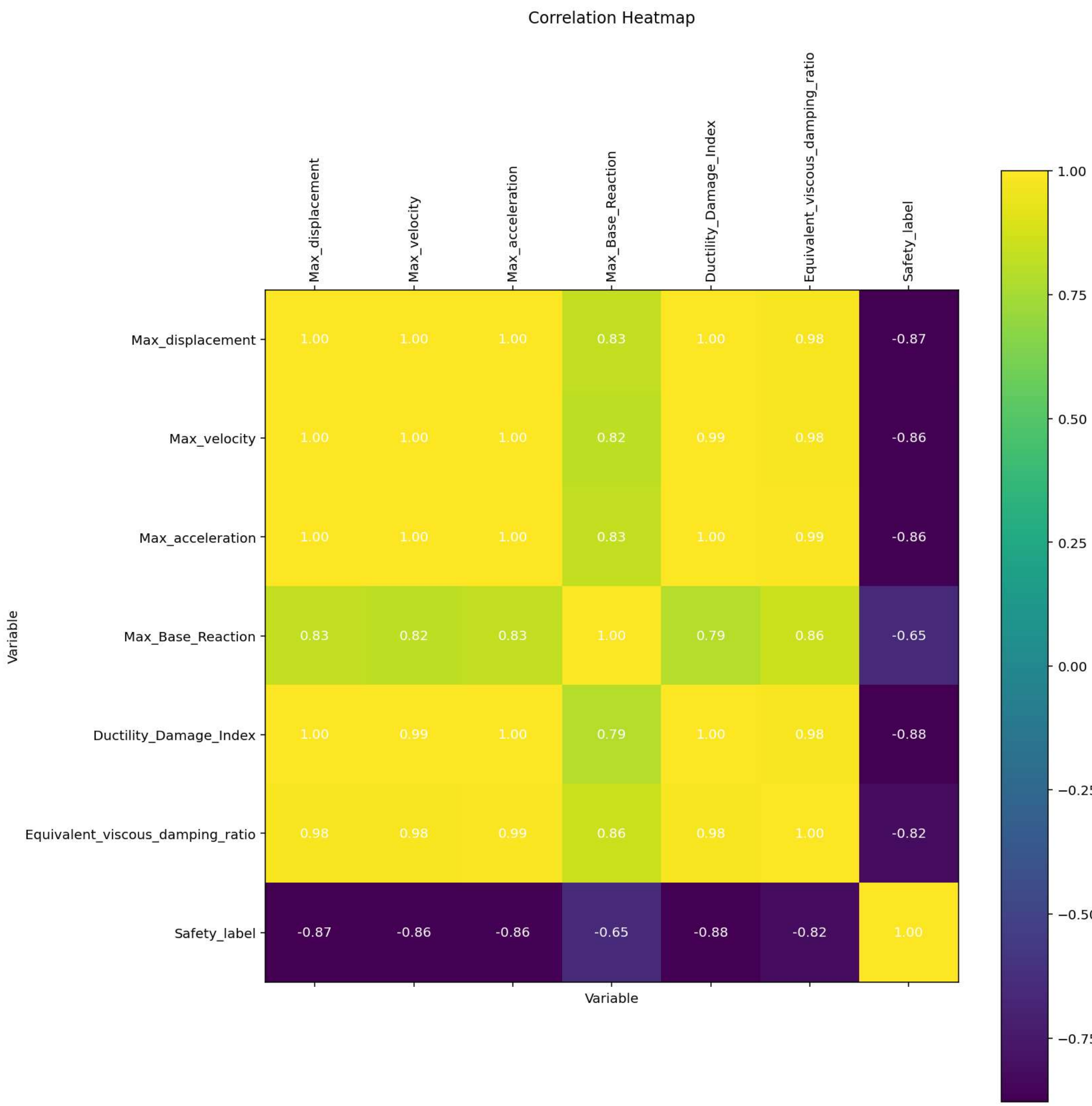
Time vs Acceleration - MAX. ABS: 8.425882924260652
Acceleration [m/s^2]
Time [s]
8
6
4
2
0
−2
−4
−6
0
2
4
6
8
10
12
14


Time vs Base-reaction - MAX. ABS: 103498.3888288074
100000
50000
0
−50000
−100000
Base-reaction [N]
0
2
4
6
8
10
12
14
Time [s]


Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
Base Reaction [N]
Time [s]



Feature Importance
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Equivalent_viscous_damping_ratio
None
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Equivalent_viscous_damping_ratio
Safety_label
Variable
1.00 1.00 1.00 0.83 1.00 0.98 -0.87
1.00 1.00 1.00 0.82 0.99 0.98 -0.86
1.00 1.00 1.00 0.83 1.00 0.99 -0.86
0.83 0.82 0.83 1.00 0.79 0.86 -0.65
1.00 0.99 1.00 0.79 1.00 0.98 -0.88
0.98 0.98 0.99 0.86 0.98 1.00 -0.82
-0.87 -0.86 -0.86 -0.65 -0.88 -0.82 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75